AADK Task 1 Report

Android Fundamentals: Learn by Doing

Name: Guru K R

Course Title: Android App Development: First Steps

Date: February 16, 2026

Contents

1 Discovering the Android Ecosystem 2

2 Visualizing the Android World 3

3 Accessibility Reflection 4

4 Critical Reflection 4

Discovering the Android Ecosystem

1.1 Android-Powered Devices

1. Smartphones

Android smartphones are the most widely used Android devices globally. They support communication, entertainment, productivity, and daily utilities.

2. Wearables (Smartwatches)

Wearables running Wear OS help users track health metrics, receive notifications, and interact hands-free.

3. Smart TVs (Android TV)

Android-powered smart TVs provide streaming, gaming, and voice-assisted browsing experiences on large screens.

1.2 Popular Android Apps and Their Impact

Google Maps (Navigation)

Problem Solved: Helps users find directions, avoid traffic, and explore nearby places.
User-Friendly Features: Real-time traffic updates, voice navigation, and intuitive interface.

Fitbit (Health & Fitness)

Problem Solved: Tracks physical activity, sleep, and heart rate to improve overall health.
User-Friendly Features: Simple dashboard, visual health statistics, and personalized insights.

Duolingo (Learning)

Problem Solved: Makes language learning accessible and engaging for users worldwide.
User-Friendly Features: Gamified lessons, progress tracking, and bite-sized learning modules.

Visualizing the Android World

Android Ecosystem Infographic

Interesting Fact:

Android powers over 3 billion active devices worldwide and supports more than 100 languages, making it the most widely used mobile operating system globally.

Accessibility Reflection

3.1 Android Accessibility Features

TalkBack
A screen reader that provides spoken feedback, helping visually impaired users navigate their devices.

Color Correction & Magnification
Allows users with visual impairments to adjust color settings and zoom into screen content for better visibility.

3.2 My Accessibility Idea

For my future Android app, I would implement **AI-powered voice navigation with simplified mode**. This feature would allow users with limited technical knowledge or disabilities to control the app entirely using voice commands and access a minimal, distraction-free interface.

Critical Reflection

Why Developers Should Design with Accessibility in Mind

Designing with accessibility ensures that applications can be used by everyone, regardless of physical or cognitive limitations. Inclusive design expands the app's reach to a broader audience and promotes equal digital participation. It also enhances overall usability, making apps simpler and more intuitive for all users. By prioritizing accessibility, developers contribute to a more inclusive and socially responsible digital ecosystem.

— End of Report —

AADK Task 2 Report

Understanding Android Studio and App Structure

Name: Guru K R

Course Title: Android App Development: First Steps

Date: February 16, 2026

Contents

1 Navigating Android Studio 2

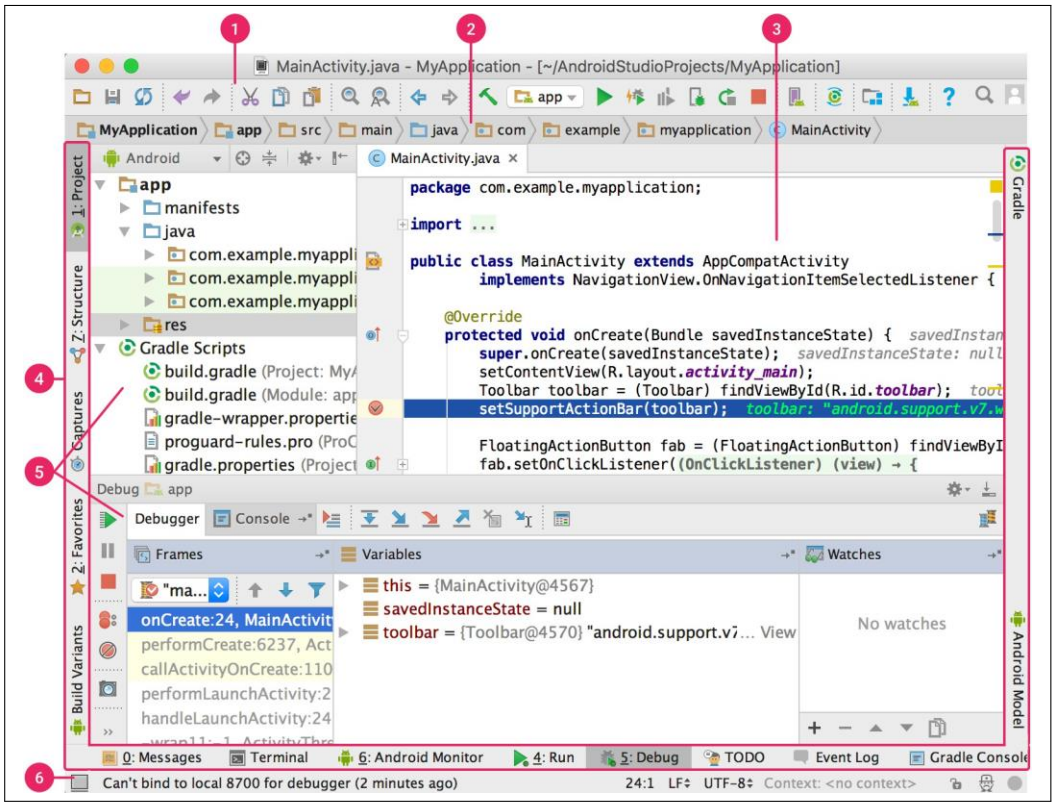
2 Exploring Project Structure 2

3 File Mapping Diagram 3

4 Mini Reflection 4

Navigating Android Studio

1.1 Android Studio Interface Screenshot



Labeled Sections:

- **Project Panel** – Displays project files, folders, and resources.
- **Code Editor** – Area where Kotlin or Jetpack Compose code is written.
- **Preview Area** – Shows a visual representation of UI components.

Exploring Project Structure

2.1 Key Project Files

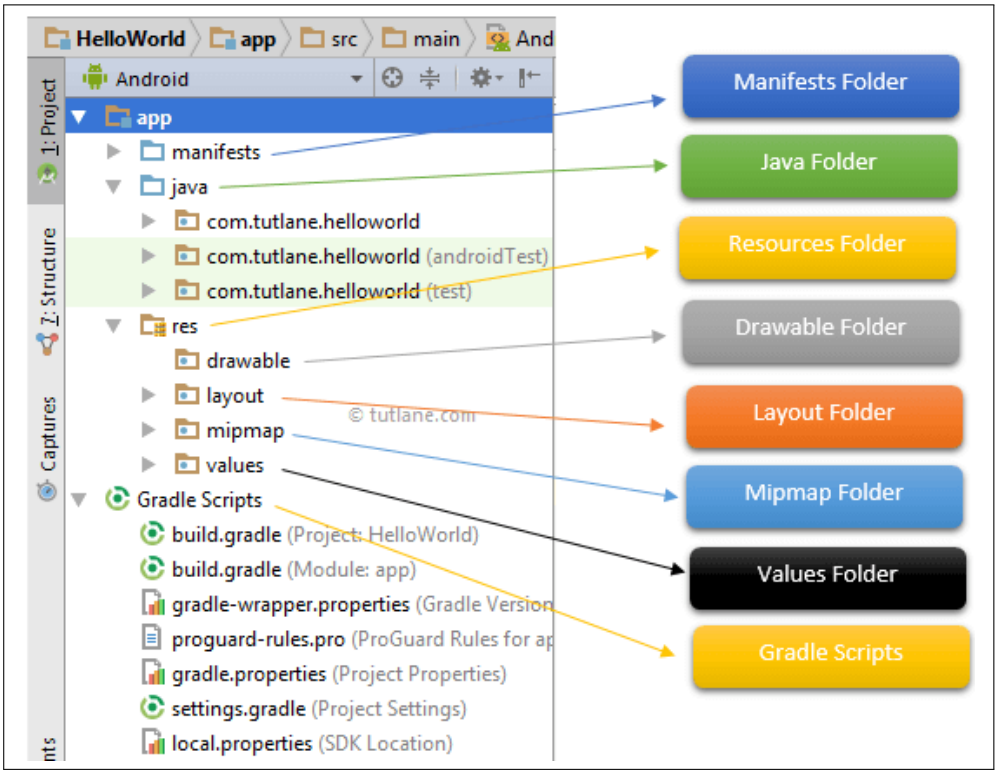
- MainActivity.kt
- AndroidManifest.xml
- res/values/strings.xml
- build.gradle (Module)

2.2 File Purpose Table

File Name	Purpose
MainActivity.kt	The main entry point of the application where app logic and UI setup begin.
AndroidManifest.xml	Contains essential app information such as permissions, activities, and app configuration.
res/values/strings.xml	Stores text strings used in the app to support easy updates and localization.
build.gradle (Module)	Defines dependencies, SDK versions, and build configuration for the app.

File Mapping Diagram

Project Structure Flow Diagram



Example Flow Explanation:

AndroidManifest.xml registers MainActivity → MainActivity loads UI resources → Resources (strings.xml) provide text content → build.gradle manages dependencies and project configuration.

Mini Reflection

Reflection

Through this task, I learned how Android Studio organizes files in a structured and logical way. Each file has a specific responsibility, from defining app configuration to managing UI resources and dependencies. Understanding this structure before coding helps developers avoid confusion and build scalable applications. It also improves debugging and collaboration, as team members can quickly locate and modify the required components.

— End of Report —

AADK Task 3 Report

Design Your First App Idea (Concept + UI Sketch)

Name: Guru K R

Course Title: Android App Development

Date: February 16, 2026

Contents

1 App Concept Creation 2

2 UI Sketch / Wireframe 3

3 Design Reflection 4

4 Optional Enhancement: Color Scheme & Typography 4

App Concept Creation

1.1 App Overview

App Name: FocusFlow

Purpose: FocusFlow helps students manage daily academic tasks efficiently by organizing assignments, deadlines, and reminders in one place.

Target Audience: College students and high school learners who want a simple productivity tool without distractions.

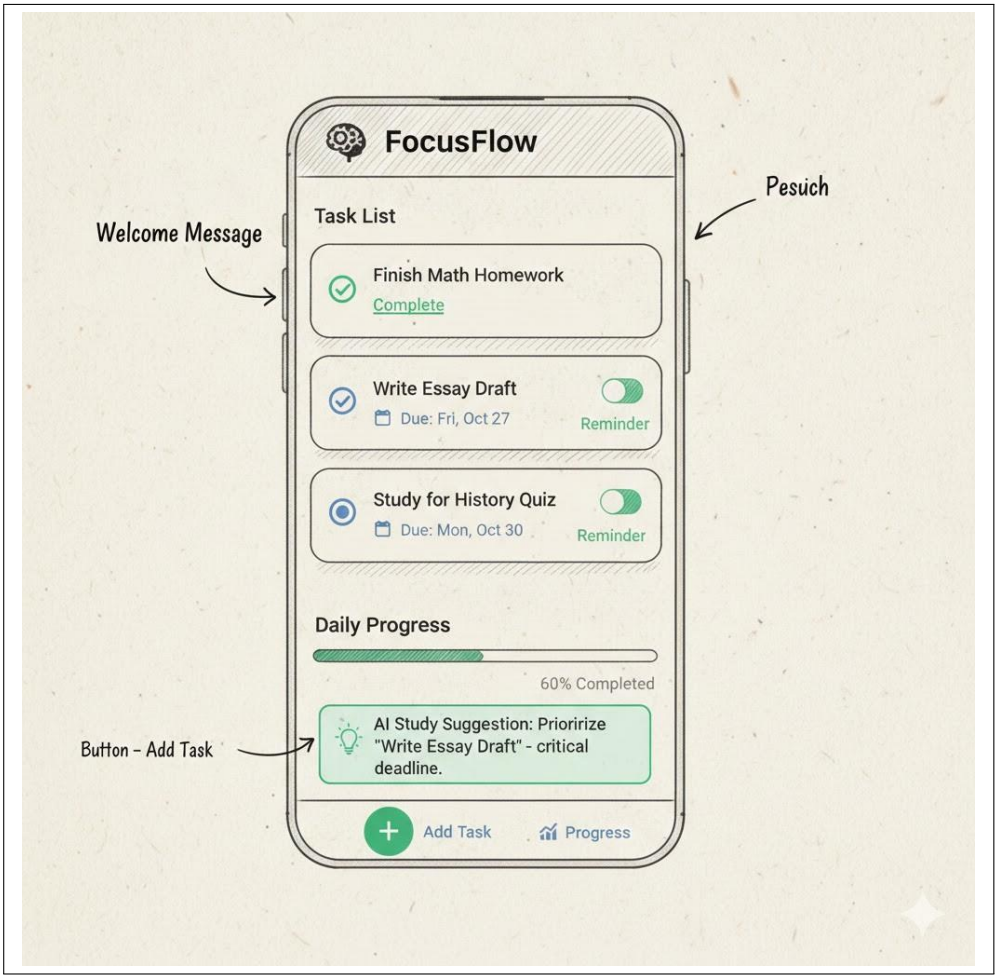
1.2 Core Features

- Task creation with deadline selection
- Daily progress tracker
- Smart reminders and notifications

Unique Feature: AI-based study suggestion that recommends task priority based on upcoming deadlines and workload.

UI Sketch / Wireframe

Main Screen Layout



UI Element Labels:

- Text – Welcome Message
- Column – Task List Layout
- Button – Add Task
- Button – View Progress
- Image – Motivational Illustration

Design Reflection

Reflection

I chose this app idea because students often struggle with organizing tasks and maintaining productivity. The design focuses on simplicity by minimizing clutter and prioritizing essential actions like adding tasks and viewing progress. A clean layout ensures ease of navigation and reduces cognitive overload. To build this interface using Jetpack Compose, I would likely use components such as **Column**, **Row**, **Text**, **Button**, and **Image**. These elements would help structure the layout efficiently while keeping the UI modern and responsive.

Optional Enhancement: Color Scheme & Typography

Design Theme

Color Scheme:

- Primary Color – Soft Blue (#4A90E2)
- Secondary Color – Light Gray (#F2F4F8)
- Accent Color – Fresh Green (#27AE60)

Typography:

- Headings – Sans-serif bold font
- Body Text – Clean and readable modern font

The selected colors aim to create a calm and focused atmosphere, enhancing productivity and readability.

— End of Report —